A Report Based on

"On Job Training"

Carried out at

BHARTI HOSPITAL AND RESEARCH CENTRE

Central Research & Publication Unit (CRPU), Pune

Submitted at

Department of Statistics,

Kaviyatri Bahinabai Chaudhari

North Maharastra University, Jalgaon

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It gives us immense pleasure to express our heartfelt gratitude to all those who played a significant role in making our internship a meaningful and enriching experience.

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We are extremely grateful to the entire team at the **Central Research & Publication Unit (CRPU), Bharati Hospital and Research Centre, Pune**, for welcoming us with warmth and providing an inspiring learning environment. Special thanks to **Dr. Rupesh Deshmukh Sir**, **Mr. Mayur Shinde Sir**, **Mr. Tejas Patil Sir**, and **Mr.Shrikant Sir** for their patient explanations, hands-on demonstrations, and continuous support during our practical training.

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This internship has been a truly transformative experience for us, and we are sincerely thankful to everyone who contributed to it in any way.

Sincerely,

INTERNS

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01 INTRODUCTION

As part of the M.Sc. Statistics curriculum at **Kaviyatri Bahinabai Chaudhari North Maharashtra University, Jalgaon**, we had the invaluable opportunity to undertake a one-month internship at **Bharati Hospital and Research Centre, Pune**, under the supervision of the **Central Research & Publication Unit (CRPU)**. This internship served as a bridge between our academic learning and its real-world applications, particularly in the field of **biostatistics, clinical research, and statistical data analysis**.

The CRPU is a dedicated unit within Bharati Hospital that facilitates clinical research activities, designs and manages medical studies, coordinates ethical review processes, and supports the use of advanced statistical methods in medical investigations. Through our engagement with the CRPU team, we were exposed to a structured and collaborative research environment where statistical knowledge is actively applied to patient care, treatment evaluation, and healthcare policy formulation.

During the internship, we were introduced to a variety of critical concepts including **sampling theory, study designs (case-control, cohort, RCTs)**, **hypothesis testing**, **ROC analysis**, and **survival models**. We also learned to use industry-relevant software such as **SPSS**, **SAS**, and open-source tools to perform descriptive and inferential statistical analyses.

In addition, we gained exposure to **machine learning techniques** such as decision trees, logistic regression, random forest, and principal component analysis (PCA), all of which have increasing relevance in modern clinical and epidemiological research. These methods were applied to real clinical datasets, allowing us to not only understand theoretical concepts but also appreciate their impact in evidence-based decision making.

Our internship was not limited to technical knowledge. It also involved developing essential research skills such as preparing study protocols, estimating sample sizes, maintaining ethical documentation, and contributing to the publication process. We also received guidance on **professional communication**, **CV preparation**, and **interview skills**, which enhanced our overall readiness for careers in clinical research, data science, and applied statistics.

Overall, this internship experience allowed us to integrate theoretical learning with practical exposure, preparing us to take on real-world statistical and research challenges with greater confidence and competence

02 Bharati Hospital and Research Centre, Pune

A Hub of Healthcare, Education, and Clinical Research

Bharati Hospital and Research Centre, located in Pune and affiliated with Bharati Vidyapeeth (Deemed to be University), stands as a well-established institution delivering both healthcare services and medical education. With a legacy of serving society since its establishment in 1989 by the visionary **Dr. Patangrao Kadam**, the hospital has evolved into a centre known for its **compassionate care, advanced technology, and educational excellence**.

Spread over 25 acres in a serene environment in Katraj, Pune, Bharati Hospital serves thousands of patients from both urban and rural regions. It offers an integrated model of service, education, and research. Its services cater to preventive, curative, diagnostic, rehabilitative, and emergency healthcare, ensuring comprehensive care under one roof.

Vision and Mission:

The hospital aims:

- To deliver **affordable, high-quality healthcare services** to all sections of society.
- To serve as a **Centre of Excellence in medical education and clinical research**.
- To foster innovation, ethical medical practices, and community-based research for better health outcomes.

Facilities and Departments:

- **Bed capacity**: Over **1,200 beds**, including general wards and specialized units like ICU, PICU, NICU, CCU
- **Diagnostic capabilities**: Equipped with **CT scans, MRI, USG, X-ray**, pathology labs, and 24x7 blood bank
- **Specialties**: Internal Medicine, General Surgery, Pediatrics, Obstetrics and Gynaecology, Psychiatry, Neurology, Oncology, Orthopaedics, Cardiology, Dermatology, and more
- **Supportive Services**: Physiotherapy, Human Milk Bank, Nutrition and Dietetics, Pediatric Rehabilitation, and Sleep Labs

Achievements and Accreditations:

- Accredited by **NABH** (National Accreditation Board for Hospitals) and **NABL** (National Accreditation Board for Testing and Calibration Laboratories)
- Recognized as a **clinical trial site** for significant research, including the **CoviShield vaccine trial**
- Accredited under **Mahatma Jyotiba Phule Jan Arogya Yojana (MJPJAY)** by the Government of Maharashtra

Bharati Hospital not only treats patients but also provides a training ground for future doctors, nurses, pharmacists, physiotherapists, and researchers. The hospital’s integration of care, teaching, and research makes it a dynamic ecosystem ideal for professional internships like ours.

03 Central Research & Publication Unit (CRPU)

Empowering Clinical Research Through Statistical Insight and Data Science

The **Central Research & Publication Unit (CRPU)** functions as the research core of Bharati Hospital. It plays a pivotal role in guiding and supervising all scientific and data-related activities. With an interdisciplinary approach that connects clinicians, statisticians, public health experts, and data scientists, the CRPU transforms raw medical observations into meaningful, publishable research.

The CRPU provided us an environment where classroom theory was translated into clinical application. It encouraged us to work on real datasets, understand ethical considerations, and appreciate the role of **evidence-based medicine** in improving healthcare delivery.

Aims and Objectives:

- To support the **design and execution of research protocols**, including interventional and observational studies
- To augment research capacity of students and faculty
- To foster impactful inter-departmental, inter=disciplinary and multicentric research
- To ensure that research complies with **ethical standards**, with appropriate documentation and patient consent
- To equip interns and staff with **analytical and software skills** needed for data processing and statistical modeling
- To facilitate **scientific writing and publication** of hospital-based research findings
- To achieve and maintain high impact factor for Bharti Vidyapeeth Medical Jourbak(BVMJ)

Scope of Work :

- Research capacity building
- Conduct and facilitate intramural and extramural research
- Guidence for publication
- Publication of peer reviewed, open access, quarterly journal BVMJ

Key Roles and Activities:

- Performing **sample size calculations**, data cleaning, coding, and management
- Conducting **biostatistical analysis** using SPSS, SAS, and R: including t-tests, chi-square, ANOVA, regression, ROC analysis, and survival analysis
- Guiding **machine learning applications** such as logistic regression, decision trees, random forest, PCA, and stepwise regression
- Coordinating with the **Institutional Ethics Committee** for protocol approvals and ensuring compliance with ethical standards
- Preparing presentations, posters, manuscripts, and helping with conference submissions

Technologies and Tools Used:

- **Statistical Software**: SPSS, SAS, R, Python
- **Sample size tools**: GitHub-based calculators
- **Documentation & Research Tools**: Microsoft Word, Excel, PowerPoint, PubMed, Zotero
- **Data Handling Tools**: SPSS, R , Excel
- **Machine Learning Models**: PCA, decision trees, logistic regression, random forest, KNN

Impact on Our Learning:

Our time at CRPU taught us how to approach data scientifically—from problem framing and protocol writing to hypothesis testing, statistical validation, and results interpretation. It bridged the gap between theoretical statistics and its life-saving potential in medicine. Most importantly, it gave us confidence in using software and analytical methods that are critical in real-world research settings.

04 Objectives of the Internship

The primary goal of our internship at the **Central Research & Publication Unit (CRPU), Bharati Hospital, Pune**, was to bridge the gap between theoretical statistical knowledge and its practical application in medical research. Through structured learning, guided mentorship, and hands-on data analysis, the internship aimed to equip us with the skills and experience required to contribute effectively to real-world healthcare studies.

The specific objectives were as follows:

1. **To understand the organizational structure and research role of CRPU in clinical and academic settings within a multidisciplinary hospital environment.**
2. **To learn the principles of study design**, including case-control, cohort, cross-sectional studies, and randomized controlled trials (RCTs).
3. **To perform sample size calculations** for various study types using appropriate statistical methods and software tools.
4. **To apply biostatistical techniques** such as descriptive statistics, hypothesis testing (parametric and non-parametric), ANCOVA, odds ratio analysis, and ROC/AUC analysis.
5. **To gain hands-on experience with statistical software** including **SPSS** and **SAS** for data management, statistical analysis, and visualization.
6. **To understand and apply survival analysis methods**, including Kaplan–Meier curves, log-rank tests, and Cox proportional hazards models.
7. **To explore the fundamentals of machine learning** in clinical data analysis, including model building, performance evaluation, and feature selection techniques like PCA and stepwise regression.
8. **To develop professional skills**, including research documentation, ethics compliance, preparation of informed consent forms, and communication with research committees.
9. **To prepare for future career roles** by improving resume writing, technical interview readiness, and scientific communication.

05 Biostatistics: Concepts and Applications

What is Biostatistics?

Biostatistics is the application of statistical principles and methods to address problems in biology, medicine, public health, and allied health sciences. It enables researchers to **transform raw medical data into meaningful evidence**, supporting healthcare decisions and policy-making.

Biostatistics combines the fields of **mathematics, statistics, and life sciences** to ensure that medical research is conducted in a scientifically valid, ethical, and interpretable way.

Core Areas of Biostatistics:

Core Area	Description	Example/Tools
1. Study Design & Planning	Designing valid medical studies (RCTs, cohort, etc.), selecting controls, randomization, and blinding.	RCTs, Blinding, Ethics Approval
2. Data Collection & Management	Organizing, cleaning, and validating data for accuracy and consistency.	Excel, REDCap, Data dictionaries
3. Descriptive Statistics	Summarizing and visualizing data to understand distribution and trends.	Mean, Median, SD, Histograms, Boxplots
4. Inferential Statistics	Drawing conclusions about populations using sample data through hypothesis testing.	t-test, ANOVA, chi-square, p-values, CI
5. Regression & Prediction Models	Exploring relationships and making predictions using mathematical models.	Linear/Logistic Regression, Poisson models
6. Survival Analysis	Analyzing time-to-event data (like time to death, recovery).	Kaplan–Meier, Log-rank test, Cox regression
7. Diagnostic Test Evaluation	Evaluating accuracy of tests to classify disease correctly.	Sensitivity, Specificity, ROC, AUC
8. Machine Learning Integration	Using advanced models to detect patterns and improve predictive performance.	Decision Trees, Random Forest, PCA, Lasso

Role of a Biostatistician in a Medical Research Team:

- Designs studies with minimal bias and optimal sample sizes.
- Advises on data collection methods and ethical issues.
- Applies the right statistical tests and validates model assumptions.
- Interprets complex results for clinicians in a meaningful way.
- Contributes to scientific writing, publication, and presentation of results.

Biostatistics is the **heart of medical research**. It ensures that conclusions drawn from data are not only statistically sound but clinically meaningful. From formulating research questions to interpreting results and publishing findings, biostatistics guides every step of the scientific research process.

06 Sampling Techniques and Sample Size Calculation

In research, sampling refers to the process of selecting a **subset (sample)** from a larger group (population) for the purpose of making inferences about the whole. It is especially essential in medical and public health research, where studying the entire population is impractical due to time, cost, or logistics.

- **Population (N):** The total group under study.
- **Sample (n):** A smaller representative group drawn from the population.
- **Sampling Frame:** A list or structure that defines the population from which the sample is drawn (e.g., voter lists, hospital records).

Non-Probability Sampling :

Method	Key Feature	Example
Convenience Sampling	Choosing what's easily available	Surveying students passing by at campus gate
Purposive Sampling	Judgement-based participant selection	Selecting cancer patients for a specific trial
Snowball Sampling	Used for rare/hidden populations	Drug abuse survey via referrals
Quota Sampling	Selection based on predefined traits	Ensuring 40% males, 60% females in a sample of 100

The **sample size (n)** depends on:

- Type of **outcome variable** (proportion or mean)
- **Confidence level** (commonly 95%)
- **Power** of the test (commonly 80%)
- **Effect size / Precision** expected (d)
- **Prevalence / baseline values** from prior studies

Test/Condition	Formula
Estimate a Single Proportion	$n = \frac{[Z_{(1-\alpha/2)}]^2 \times P \times (100-P)}{d^2}$
Compare Two Independent Proportions	$n = \frac{[Z_{(1-\alpha/2)} + Z_{(1-\beta)}]^2 \times [p_1(100-p_1)+p_2(100-p_2)]}{(p_1-p_2)^2}$
Estimate a Single Mean	$n = \frac{[Z_{(1-\alpha/2)}]^2 \times (SD)^2}{d^2}$
Compare Two Independent Means	$n = \frac{2 \times [Z_{(1-\alpha/2)} + Z_{(1-\beta)}]^2 \times (pooledSD)^2}{(M_1-M_2)^2}$
Diagnostic Accuracy (Sensitivity)	$n = \frac{[Z_{(1-\alpha/2)}]^2 \times SN \times (1-SN)}{d^2 \times Prevalence}$
Diagnostic Accuracy (Specificity)	$n = \frac{[Z_{(1-\alpha/2)}]^2 \times Sp \times (1-Sp)}{d^2 \times (1-Prevalence)}$

Notation:

- $Z(1-\alpha/2)$): Z-score for confidence level (typically 1.96 for 95%)
- $Z(1-\beta)$*:* Z-score for power (typically 0.84 for 80%)
- *P*: Prevalence or proportion (%)
- d: Allowable error (absolute or relative difference)
- p1,p2: Proportions in group 1 and 2 (%)
- SD: Standard deviation
- M1,M2: Means of group 1 and 2
- pooledSD: Pooled standard deviation
- SN: Sensitivity
- Sp: Specificity
- Prevalence: Disease prevalence

Sample Size Calculated using GitHub Sample Size Calculator [Link] –

https://www.google.com/search?q=github+sample+size+calculator&oq=githu&gs_lcrp=EgZjaHJvbWUqDAgCECMYJxiABBiKBTIGCAAQRRg5MhMIARAuGIMBGMcBGLEDGNEDGIAEMgwIAhAjGCcYgAQYigUyBggDEEUYOzINCAQQABiDARixAxiABDIKCAUQABixAxiABDIKCAYQABixAxiABDIKCAcQABixAxiABDIHCAgQABiPAtIBCTQ2NjNqMGoxNagCCLACAfEFyPHluq2dJzA&sourceid=chrome&ie=UTF-

GITHUB SAMPLE SIZE CALCULATOR (SINGLE PROPORTION) :

Sample Size Calculator (web)

1 proportion - Estimation

Proportion (p):

0.84

Precision (± proportion):

0.084

Confidence level 100(1 - α):

95 %

Expected dropout rate:

10 %

Calculate

Reset

Sample size, n =

74

Sample size (with 10% dropout), n_{drop} =

83

Biostatistics Unit

Date: 27/05/2025

Format for approval of study design and sample size estimation of the Study/Project/Thesis

1	**Title of the Study/Project /Thesis**	Effectiveness of a structured ergonomic training and micro-streching programme on work-related Musculoskeletal Disorders among Intensive care unit Nurses : A pretest-post-test study
2	**Department Name**	Physiology
3	**Name of the Student/Researcher**	Priyanka Suyash Limaye
4	**Guide/PI Name**	Dr. Gayatri Godbole

		Observational		Interventional/ Experimental	Diagnostic/ Predictive Study
5	**Type of Study Design** (Tick ✓ on appropriate design)	**Descriptive**	**Analytical**		
		Cross-sectional	Cross-sectional	RCT	
			Case-control	Quasi-experimental	
			Cohort study	Pre-Post ✓	
			Ambidirectional	Community Trial	

For estimation of sample size

6	**Primary Objective of the Study /Project/Thesis**	To estimate prevalence of work-related Musculoskeletal Disorders among Intensive care Unit nurses		
7	**Estimates from appropriate reference:** Prevalence/Proportion/ Mean/SD/Difference between two groups/ Sensitivity/Specificity/ Allowable error (d) etc.	The prevalence of work-related musculoskeletal Disorders in Intensive Care Unit nurses = 84 %. Difference d = 8.4 (10 % of prevalence)		
8	**Citation of the reference for point no. 7**	comprehensive investigation of ergonomic challenges and predictors of work-related musculoskeletal disorders among intensive care unit nurses of western India through convergent mixed method study Sonali Detroja'		
9	**Power of the test (1-B)**	80%	10 **Confidence Interval**	95%
11	**Estimated Sample Size**	n = 73		

Declaration by Guide and Student/Researcher

I hereby declare that I have **read** the article cited above and am fully aware of the estimates used for the sample size calculation. As a PG guide, I understand my responsibility to ensure the accuracy and appropriateness of these estimates for the study's primary objective. I acknowledge that it is essential to carefully review and verify all aspects of the sample size calculation to ensure the integrity and validity of the research.

ignature of the Student Researcher: Limaye

ignature of the guide: ______

Deshmukh 27/5/25

Signature & Name of the Biostatistician

Dr. Rupeshkumar Deshmukh
Assistant Professor cum Biostatistician
...Medicine

07 Study Designs in Medical Research

What is a Study Design?

In the field of biostatistics and clinical research, a **study design** is the detailed plan or framework used to conduct a scientific investigation. It outlines **how data will be collected, from whom, for how long, and in what manner**—all in alignment with the study's primary objective.

A well-chosen study design ensures that:

- The **data collected are valid and reliable**
- The **results can be interpreted statistically**
- Bias is minimized, and **causal relationships** (if any) can be explored

Study designs form the foundation of research methodology and influence everything from **sample size estimation**, **choice of statistical tests**, to the **strength of the conclusions drawn**.

Type	Subtypes
Observational	Descriptive (e.g., cross-sectional), Analytical (e.g., case-control, cohort)
Experimental	Interventional (e.g., randomized controlled trials - RCTs)

When to Use Which Study Design?

Objective	Best Study Design
Estimate disease prevalence	**Cross-sectional Study**
Explore rare disease risk factor	**Case-Control Study**
Investigate cause-effect over time	**Cohort Study**
Test treatment/intervention	**Randomized Controlled Trial**

1. Cross-Sectional Study:

Aspect	Description
Definition	Observes a population at a single point in time to assess health status or conditions.
Time Direction	None (snapshot study)
Data Collected	Prevalence of diseases, exposures, risk factors
Measure of Effect	Prevalence Rate
Advantages	Fast, inexpensive, useful for public health planning
Limitations	Cannot establish cause-effect or temporal relationships
Example	Survey of diabetes prevalence among adults in a city

2. Case-Control Study:

Aspect	Description
Definition	Compares patients with a disease (cases) to those without (controls) to identify past exposures.
Time Direction	Retrospective (looks backward in time)
Data Collected	Exposure history, risk factor presence
Measure of Effect	Odds Ratio (OR)
Advantages	Suitable for rare diseases, cost-effective
Limitations	Recall bias, can't calculate incidence or establish causality
Example	Comparing smoking history between lung cancer patients (cases) and healthy people (controls)

3. Cohort Study:

Aspect	Description
Definition	Follows a group over time to compare disease incidence between exposed and unexposed groups.
Time Direction	Prospective or retrospective
Data Collected	Exposure status, time-to-event, outcome incidence
Measure of Effect	Relative Risk (RR)
Advantages	Can study multiple outcomes, temporal relationships, incidence
Limitations	Expensive, time-consuming, loss to follow-up
Example	Following smokers and non-smokers over 10 years to compare heart disease incidence

4. Randomized Controlled Trial (RCT):

Aspect	Description
Definition	Participants are randomly assigned to intervention or control groups to test a treatment.
Time Direction	Prospective
Data Collected	Outcome measures post-intervention (e.g., BP reduction, cure rates)
Measure of Effect	Treatment effect, Relative Risk (RR), Absolute Risk Reduction (ARR)
Advantages	Gold standard for evaluating interventions, minimizes bias
Limitations	Costly, ethical challenges, strict eligibility criteria
Example	Testing a new vaccine by comparing outcomes in vaccinated vs. placebo group

08 Statistical Software Used: SPSS and SAS

Why Statistical Software Matters in Biostatistics

In medical research, accurate statistical analysis is crucial for making reliable conclusions, whether it's comparing treatment groups, evaluating test accuracy, or studying disease patterns. Software tools like SPSS and SAS automate calculations, reduce human error, and make large datasets manageable. They are vital for:

- Cleaning and transforming clinical data
- Running statistical tests (e.g., t-tests, chi-square)
- Visualizing results (e.g., ROC curves, bar plots)
- Interpreting data for reports and publications

SPSS – Statistical Package for the Social Sciences

Primary software used during the internship

SPSS is a powerful, user-friendly software developed by IBM used for performing statistical analysis in fields like medicine, public health, psychology, sociology, and education. It helps researchers manage, analyze, and visualize data without needing to write complex code.

It is widely used in clinical trials, hospital-based research, health surveys, and social science studies — especially when the user is more comfortable with click-based navigation rather than programming.

Key Components of SPSS

Component	Description
Data View	Like an Excel sheet – you enter rows (cases) and columns (variables)
Variable View	Defines each variable (name, type, label, values, missing codes, etc.)
Output Viewer	Shows the results of your statistical tests, tables, and charts
Syntax Editor	Optional – lets you write code to automate or repeat analyses
Chart Editor	Helps customize graphs and export them to Word or PowerPoint

SPSS Functioning:

Function	Example/Use in Medical Research
Data Entry & Cleaning	Enter patient records, remove missing values, create groups (e.g. age)
Descriptive Statistics	Find mean, median, standard deviation of blood pressure or sugar levels
Cross-tabulations	Compare gender vs disease presence using a contingency table
Charts and Graphs	Create histograms, boxplots, pie charts for presentations
T-tests and ANOVA	Compare blood pressure between treatment groups
Chi-square Test	Test association between smoking and lung disease
Correlation Analysis	Find the strength of relationship between age and cholesterol
Linear/Logistic Regression	Predict blood pressure from age, or diabetes from obesity (binary outcome)
ROC Curve & AUC	Evaluate how good a diagnostic test is
Non-parametric Tests	When data doesn't follow normal distribution (e.g., Mann–Whitney test)

Tasks Performed with SPSS During Internship:

- Imported patient datasets from Excel/CSV
- Recoded variables and created new categorical variables
- Performed descriptive analysis (mean, SD, frequency tables)
- Conducted t-tests, chi-square tests, ANOVA, and logistic regression
- Drew ROC curves and calculated AUC to evaluate diagnostic accuracy
- Interpreted outputs for protocol reports and publications

Why SPSS was Ideal for Medical Research:

- Easy to understand for clinicians and researchers with no programming background
- Quick to perform hypothesis tests and generate summary tables
- Built-in output viewer helps create direct reporting material for manuscripts

Advantages of SPSS:

User-friendly interface (menu-driven, no coding required)

Handles large datasets easily

Excellent for both basic and advanced statistical analysis

Produces professional-quality charts, graphs, and tables

Trusted by researchers, students, and organizations worldwide

SAS – Statistical Analysis System

Secondary tool used during internship

SAS is a command-based statistical software preferred in large-scale clinical trials, regulatory submissions, and advanced modeling. It's known for its robust performance, especially when dealing with large medical datasets.

Feature	Description
Programming Language	Uses DATA step and PROC step syntax
Advanced Procedures	Survival analysis (PROC LIFETEST), regression (PROC LOGISTIC/GLM)
Data Handling	Can handle millions of records efficiently
Regulatory Acceptance	Widely used in pharmaceutical industries and accepted by **FDA**

A typical SAS program has three key parts:

1. **DATA Step** – for reading in and preparing data.
2. **PROC Step** – for performing statistical procedures and tasks.
3. **Output** – for creating results in tables, charts, or summary reports.

09 Survival Analysis

Survival analysis is a branch of statistics that deals with analyzing the time duration until the occurrence of a particular event such as death, relapse, recovery, or hospital discharge. It is particularly important in medical research where not all individuals may experience the event during the study period — leading to censored observations.

During our internship at CRPU, Bharati Hospital, we were exposed to real patient datasets and guided in the application of various survival analysis techniques to assess treatment outcomes, disease progression, and model-based prediction of survival times.

We applied techniques such as:

- Kaplan–Meier Estimator
- Log-Rank Test
- Cox Proportional Hazards Model
- ROC Curve Analysis

These techniques allowed us to study survival patterns, compare treatment effects, and evaluate diagnostic or predictive models.

Kaplan–Meier Survival Curve :

The Kaplan–Meier (KM) method is a non-parametric estimator used to estimate the survival function from lifetime data. It allows the visualization of how many patients survive over a period of time, accounting for censored data (e.g., patient lost to follow-up).

Term	Explanation
Survival Function S(t)	Probability that a patient survives longer than time t
Median Survival Time	Time at which 50% of patients have experienced the event
Censored Data	Patients who didn't experience the event during study period

Internship Use :

We used KM curves to compare time-to-death between treated vs untreated groups.

- ### Log-Rank Test :

The Log-Rank Test is used to compare two or more Kaplan–Meier survival curves to determine if there's a significant difference in survival between groups.

Term	Explanation
Null Hypothesis	Survival experience is the same across groups
p-value	If $p < 0.05$, we reject the null hypothesis → survival difference is statistically significant.

Internship Use: We used this to test if there's a significant difference in survival between treatment groups.

- ### Cox Proportional Hazards Model :

The Cox model is a regression-based survival method that helps evaluate how multiple factors (age, comorbidities, treatment) affect survival time.

Term	Explanation
Hazard Function h(t)	Instantaneous event rate at time t, given survival up to t
Hazard Ratio (HR)	The ratio of hazards between two groups: - HR > 1 → higher risk - HR < 1 → lower risk
Confidence Interval (CI)	If 95% CI does not include 1 → result is statistically significant

- ### ROC Curve (Receiver Operating Characteristic) :

ROC curve is a graphical method used to evaluate the performance of a binary classification test — especially diagnostic tests in medicine.

It shows how well a test can distinguish between two groups:

- Disease vs No disease
- Positive vs Negative outcome

Measure	Explanation
Sensitivity (True Positive Rate)	% of actual positives correctly identified (e.g., sick patients diagnosed as sick)
Specificity (True Negative Rate)	% of actual negatives correctly identified
Cut-off Point	Value above which a test is positive
AUC (Area Under Curve)	Probability that the model ranks a randomly chosen positive higher than a negative case.

ROC curve plots: Y-axis: Sensitivity

X-axis: 1 – Specificity

Each point represents a different cut-off of the test :

AUC Value	Interpretation
1.0	Perfect test
0.9 – 1.0	Excellent discrimination
0.8 – 0.9	Good
0.7 – 0.8	Fair
0.6 – 0.7	Poor
0.5	No discrimination (random guess)

Internship Use: We generated ROC curves for logistic regression models and diagnostic indicators to evaluate their classification accuracy. AUC values > 0.8 indicated good performance.

- Survival analysis allows us to understand time-to-event patterns, account for censoring, and model risk over time.
- ROC curves help determine how well a diagnostic test or model performs, using sensitivity, specificity, and AUC as core metrics.
- Both tools are cornerstones of biostatistics in clinical trials, hospital studies, and epidemiological research.

Key Concepts in Survival Analysis :

Term	Meaning
Survival Time (T)	Time from a defined starting point (e.g., diagnosis) to the occurrence of the event (e.g., death)
Censoring	When we don't observe the event within the study period (e.g., patient left or is still alive)
Survival Function (S(t))	Probability that a subject survives longer than time t ($S(t) = P(T > t)$)
Hazard Function (h(t))	The instantaneous risk of the event occurring at time t (given survival until t)

10 ASSIGNMENT (PYTHON)

Logistic Regression

1. Introduction Logistic Regression is a statistical method used for analyzing datasets in which the outcome variable is binary or categorical. Unlike linear regression, which predicts continuous outputs, logistic regression is used to estimate the probability of a binary response based on one or more predictor variables. It is widely used in fields such as medical research, social sciences, and marketing.

2. Purpose of Logistic Regression The main objective of logistic regression is to model the probability that a given input belongs to a particular category. For binary classification problems, it predicts whether an event occurs (Yes/No, 1/0, Success/Failure).

3. Logistic Function (Sigmoid Function) The logistic regression model uses the logistic function to constrain predictions between 0 and 1.

Where:

is the predicted probability

is the intercept

are coefficients

are the independent variables

4. Log-Odds (Logit Model) Logistic regression models the log-odds of the probability of the dependent variable: This logit transformation ensures a linear relationship between predictors and the log-odds.

5. Types of Logistic Regression:

Binary Logistic Regression: Two possible outcomes (e.g., disease = yes or no)

Multinomial Logistic Regression: More than two unordered categories (e.g., choosing between types of transport)

Ordinal Logistic Regression: More than two ordered categories (e.g., rating scale: poor, fair, good)

6. Assumptions of Logistic Regression :

The dependent variable is binary or categorical

Observations are independent

No multicollinearity among predictors

Linear relationship between independent variables and log-odds

Large sample size for stability

7. Model Evaluation Metrics

Accuracy

Precision, Recall, F1 Score

ROC Curve and AUC

Confusion Matrix

Pseudo R-squared (e.g., McFadden's R^2)

8. Output Interpretation Model coefficients are interpreted using odds ratios (OR =). For example:

OR = 1: No effect

OR > 1: Positive association

OR < 1: Negative association

Example: If OR = 1.5 for age, a one-unit increase in age increases the odds of the outcome by 50%.

9. Applications of Logistic Regression

Disease diagnosis (e.g., predicting diabetes)

Email spam detection Customer churn prediction

Credit scoring

Risk assessment

10. Logistic Regression in SPSS, R, and Python

SPSS: Analyze > Regression > Binary Logistic

R: glm(outcome ~ predictors, family = binomial)

Python: from sklearn.linear_model import LogisticRegression

2. **Conclusion** Logistic regression is a fundamental classification algorithm that is easy to implement and interpret. Its applications in healthcare, business, and social science research make it an essential tool for statisticians and data scientists. Although limited in handling nonlinear

relationships and complex datasets, logistic regression remains a reliable choice for many binary classification problems.

DECISION TREE

What is decision tree?

A decision tree is a type of supervised machine learning model used for both classification (predicting categories) and regression (predicting numerical values). It works much like a flowchart:

How is it work?

1. splitting – At each internal node, the model selects the feature and threshold that best separates the data. This is based on measures like information gain or Gini impurity.

2. Recursive partitioning – This process continues down each branch, splitting subsets further until a stopping condition is met (e.g., max depth, minimum samples).

3. Prediction – For a new input, you follow the tree's conditions from the root to a leaf, which gives the final decision**.**

steps for building a decision tree (classification or regression):

o Data Preparation

o Split Dataset (train/test)

o Select Root Feature using criteria (Gini, Entropy, MSE)

o Split Data based on the selected feature

o Recursive Splitting on each branch

o Check Stopping Criteria

o Assign Leaf Output

o Prune the Tree.

o Evaluate Model using test data (accuracy, precision, RMSE, etc.)

o Make Predictions by traversing from root to leaf on new samples

Why we use decision tree?

Highly interpretable

Works with both numerical & categorical data

Handles missing values and outliers

Captures non-linear relationships

Requires minimal data pre-processing

Built-in feature selection & importance ranking

Fast to train and predict

Limitations of Decision Trees (Short Summary)

- Overfitting – Trees can become too complex and fit noise in the data.
- High Variance – Small changes in data can lead to very different trees.
- Bias Toward Features with Many Categories – May prefer variables with more levels.
- Poor for Linear Patterns – Doesn't model smooth linear relationships well.
- Instability – Results may vary with different data samples.
- Prone to Errors in Sparse Data – If data is small or imbalanced, predictions may be unreliable.
- Interpretability Drops in Deep Trees – Large trees become hard to understand.

Random Forest Classifier

Random Forest is a powerful and widely used ensemble machine learning algorithm for classification and regression tasks. It belongs to the family of bagging (Bootstrap Aggregation) techniques and is designed to improve the performance of individual decision trees by combining multiple trees into a single, stronger model.

Working Steps of Random Forest Classifier:

1. Bootstrap Sampling: Create multiple datasets by randomly sampling the original dataset with replacement.

2. Train Decision Trees: Train a separate decision tree on each bootstrap sample. At each node, use a random subset of features to determine the best split.

3. Aggregate Results:

o For classification: Each tree gives a predicted class, and the majority vote is chosen as the final prediction.

o For regression: The average of predictions is taken.

Recursive Feature Elimination (RFE)

Recursive Feature Elimination (RFE) is a popular technique in machine learning for selecting the most important features in a dataset. It is especially useful when there are many predictors, and the goal is to reduce dimensionality while maintaining model accuracy.

Key Concept: RFE works by recursively fitting the model, ranking features by importance, and removing the least significant features step-by-step.

Steps in RFE:

1. Train the model (e.g., Logistic Regression, Random Forest).
2. Rank features based on their importance.
3. Remove the least important feature.
4. Repeat until the desired number of features is reached.

Applications of RFE:

Enhancing model accuracy

Reducing overfitting

Improving model interpretability

When to Use RFE:

When you have a large number of variables

When some variables are irrelevant or redundant

In pre-processing before fitting models like logistic regression or decision tree

Stepwise Regression

Variable Selection Techniques

Stepwise regression is a systematic method of adding or removing predictors from a multiple regression model based on their statistical significance. The goal is to build a model that includes only the most meaningful predictors, improving model simplicity and accuracy.

It is useful when:

There are many potential predictor variables.

You want to identify the most influential ones.

You aim to avoid overfitting.

Stepwise Selection (Both Directions)

A combination of forward selection and backward elimination. Variables can be added or removed at each step, based on their contribution

Steps:

1. Start with no variables or a partial model.

2. Add the most significant variable.

3. Check if any of the already included variables have become non-significant.

4. Remove the non-significant ones.

5. Repeat until the best set of predictors is found.

Limitations of Stepwise Regression

Can miss combinations of variables that work well together.

Based purely on statistical significance, not domain knowledge.

Can lead to overfitting if not validated with test data.

Multicollinearity can affect variable selection.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a powerful dimensionality reduction technique used in statistics and machine learning to simplify complex datasets. It helps in identifying the most important patterns (components) in the data while retaining as much information (variance) as possible.

Purpose of PCA

In medical and biostatistical research, PCA is often used when:

- Multiple variables are highly correlated (multicollinearity)
- The dataset contains redundant or overlapping information
- We want to reduce the number of variables for easier interpretation or modelling

How PCA Works

PCA transforms the original correlated variables into a new set of uncorrelated variables called principal components (PCs). These components are:

- Linear combinations of the original variables
- Ordered by the amount of variance they explain:
  - PC1 = component that explains the most variance
  - PC2 = second most, and so on

Steps Involved in PCA

1. Standardize the data (mean = 0, SD = 1)
2. Compute the covariance matrix
3. Extract eigenvalues and eigenvectors
4. Sort and select top components that explain most variance
5. Project the original data onto these components

Benefits of PCA

- Reduces dimensionality of the data
- Solves the multicollinearity problem in regression
- Improves model performance in some cases
- Helps visualize data in 2D or 3D using top components

Conclusion :

PCA is an essential tool in biostatistics when dealing with high-dimensional and correlated data. During our internship, PCA proved especially useful in preprocessing clinical datasets before applying machine learning and regression models.

Assignment -

https://drive.google.com/file/d/1MEZudYkPGuIRzGfAeg4ptaSHM5gmEXZD/view?usp=drive_link

11 YSMC Dataset Analysis using SPSS

https://docs.google.com/spreadsheets/d/1UwgLA_MUj9mFLrKWOu9hg74dYpMgEs5t/edit?usp=drive_link&ouid=102361808035996695223&rtpof=true&sd=true

STEPS OF ANALYSIS PERFORMED ON DATASET [SPSS # TABLE 1]

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then Demographic variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data & **Categories and Totals** for categorical data & click **ok.**
- P value obtained for Age, BMI, Duration in Months such as, Select **Analyse** -> **Compare Means** -> **Independent-Sample T Test** -> opt Study Group as **Grouping Variable** & Age, BMI, Duration in Months as **Test Variables**(s) then Define Groups as per encoding is done & Click ok.
- P value obtained for Sex such as, Select **Analyse** -> **Descriptive Statistics** -> **Crosstabs** -> opt Study Group in **Columns(s)** and categorical variable Sex in **Row(s)** -> at the right of the table select **Statistics** -> tick on **Chi-Square** & click OK.

TABLE 1		Study Group(Probiotic/Standard Group)								p value
		A(Probiotic)				B(Standard)				
		Count	Column N %	Mean	Standard Deviation	Count	Column N %	Mean	Standard Deviation	
Sex	**F**	4	13%			4	13%			**0.989**
	M	26	87%			26	87%			**0.54**
Age				49.7	8.8			49.7	9.2	**0.01**
BMI				27.2	4.3			27.9	4.4	
Duration in Months				5.1	1.5			3.9	2.0	**> 0.999**

CONCLUSION [Table 1] – Analysis based on Demographic variables.

- **From the p value > 0.999, we conclude that there is no significant association between two categorical variables Sex & Study Group.**
- **AS we can see average age between two independent groups is same & p value >0.99 at 5% l.O.S , i.e we conclude that there is no statistically significant difference between average age of two independent study groups.**
- **The same conclusion is made for BMI based on p value = 0.542 > 0.05, i.e average BMI for both study groups is not significantly different.**
- **Here in Duration of Months, based on p value = 0.012 < 0.05 , we can say that the mean duration of months is statistically significantly differ for both the study groups.**

STEPS OF ANALYSIS PERFORMED ON DATASET [SPSS # TABLE 2]

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then enter study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data & **Categories and Totals** for categorical data & click ok.
- P value obtained for **Bilirubin(mg/dl), Albumin(g/dl), INR(sec), Creatinine(mg/dl), Serum.Na, CRP** such as, Select **Analyse** -> **Compare Means** -> **Independent-Sample T Test** -> opt **Study Group(Probiotic/Standard Group** as **Grouping Variable** Definre Groups as per encoding done & Click ok.

	Study Group(Probiotic/Standard Group)				
	A(Probiotic)		B(Standard)		
TABLE 2	Mean	Standard Deviation	Mean	Standard Deviation	p value
Bilirubin(mg/dl)	5.345	3.553	5.483	3.132	**0.873**
Albumin(g/dl)	2.593	0.524	2.653	0.491	**0.649**
INR(sec)	1.526	0.356	1.558	0.340	**0.723**
Creatinine(mg/dl)	0.857	0.266	0.921	0.309	**0.393**
Serum.Na	133.233	3.491	131.833	4.061	**0.158**
CRP	32.354	31.400	29.257	26.405	**0.681**

CONCLUSION [Table 2] – **t-test for Independence has performed**

- **AS we can see clearly p value for for each study variable is greater than 5% level of significance. So we can conclude that Average levels of Bilirubin , Albumin, INR & Creatinine are same in both the Group of Patients.**
- **Similar Conclusion can be drawn for the variables Serum.Na & CRP based on p value > 5% l.o.s.**

STEPS OF ANALYSIS PERFORMED ON DATASET [SPSS # TABLE 3]

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data -> select **Count & Column Percent** from **Statistics** -> **Apply to Selection/ Apply to ALL** & click ok.
- P value obtained for all categorical study variables such as, Select **Analyse** -> **Descriptive Statistics** -> **Crosstabs** -> opt Study Group in **Columns(s)** and insert all categorical variable in **Row(s)** -> at the right of the table select **Statistics** -> tick on **Chi-Square** & click OK.

TABLE 3 UPDATED		Study Group(Probiotic/Standard Group)				p value
		A(PROBIOTIC)		B(STANDARD)		
		Count	Column N %	Count	Column N %	
Chest X Ray PA view	COPD	0.00	0.00%	1.00	3.33%	0.453
	NORMAL	20.00	66.67%	16.00	53.33%	
	PLEURAL EFFUSION	10.00	33.33%	12.00	40.00%	
	pulmonary edema	0.00	0.00%	1.00	3.33%	
Ascitic Analysis	NO ASCITES	1.00	3.33%	0.00	0.00%	0.313
	NO SBP	29.00	96.67%	30.00	100.00%	
Ascitic fluid Culture	NO GROWTH	29.00	96.67%	30.00	100.00%	NA
Blood culture	NO GROWTH	30.00	100.00%	30.00	100.00%	NA
CTP Class	B	5.00	16.67%	3.00	10.00%	0.706
	C	25.00	83.33%	27.00	90.00%	
Urine Routine	NO GROWTH	0.00	0.00%	27.00	90.00%	0
	NORMAL	30.00	100.00%	3.00	10.00%	
Urine Culture	NO GROWTH	30.00	100.00%	30.00	100.00%	NA

CONCLUSION [Table 3] – **Chi-Square test of association has performed**

- **AS we can see clearly p value for for study variables Chest X Ray PA view, Ascitic Analysis & CTP Class is greater than 5% level of significance. So we can conclude that there is no Statistically significant association between these study variable with respective groups of patient.**
- **P value is significant only for the categorical variable Urine Routine , which is < 0.05 with cramer's V value 0.905. this implies Urine Routine and Study Group are not independent on each other.**
- **NA values are shown out because study variable has only single category, which implies degrees of freedom for test statistics to become 0.**

STEPS OF ANALYSIS PERFORMED ON DATASET [SPSS # TABLE 4]

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data & **Categories and Totals** for categorical data & click ok.
- P value obtained for **CTP Score, MELD Score(Initial), MELD Score(Final)** such as, Select **Analyse** -> **Compare Means** -> **Independent-Sample T Test** -> opt **Study Group(Probiotic/Standard Group** as **Grouping Variable** Define Groups as per encoding done & Click ok.

	Study Group(Probiotic/Standard Group)				
	A(Probiotic)		B(Standard)		
TABLE 4	**Mean**	**Standard Deviation**	**Mean**	**Standard Deviation**	**p value**
CTP Score	10.4	1.3	10.6	1.0	**0.508**
MELD Score(Initial)	20.0	5.2	22.7	5.1	**0.5**
MELD Score(Final)	17.2	5.9	22.7	6.1	**0.001**

CONCLUSION [Table 4] – **T-Test for Independence has performed**

- **AS we can see clearly p value for for study variable CTP Score & Initial MELD Score is greater or equal to 5% level of significance. So we can conclude that Average CTP Score and Initial MELD Score are same in both the Group of Patients.**
- **But it is found that, average Final MELD Score is significantly differ for both the Study Groups as p value < 0.05.**

STEPS OF ANALYSIS PERFORMED ON DATASET [SPSS # TABLE 5]

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data -> select **Count & Column Percent** from **Statistics** -> **Apply to Selection/ Apply to ALL** & click ok.
- P value obtained for all categorical study variables such as, Select **Analyse** -> **Descriptive Statistics** -> **Crosstabs** -> opt Study Group in **Columns(s)** and insert **Complication** variable in **Row(s)** -> at the right of the table select **Statistics** -> tick on **Chi-Square** & click OK.

		Study Group(Probiotic/Standard Group)				
		A(Probiotic)		B(Standard)		
TABLE 5		Count	Column N %	Count	Column N %	p value
Complication	Death	1.00	3.45%	2.00	6.67%	
	Hepato - Renal syndrome	2.00	6.90%	3.00	10.00%	
	Lost to FU	1.00	3.45%	3.00	10.00%	
	NIL	19.00	65.52%	13.00	43.33%	
	Overt hepatic Encephalopathy	3.00	10.34%	4.00	13.33%	
	Spontaneous bacterial peritonitis	1.00	3.45%	3.00	10.00%	
	Variceal Bleed	2.00	6.90%	2.00	6.67%	0.706

CONCLUSION [Table 5]- **Chi-square for no association has performed.**

- **AS we can see clearly p value for for study variable named Complication is greater or equal to 5% level of significance. So we can conclude there in no association between the Complications in both the Study Groups.**

ANALYSIS PERFORMED ON DR.TANMAY DATASET

Dataset 2 Link --

https://docs.google.com/spreadsheets/d/1UOKflZ0usNHkmE1oseVn2uut2ibw8QJY/edit?usp=sharing&ouid=102361808035996695223&rtpof=true&sd=true

Steps to get Kaplan-Meier in SPSS –

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Survival** –> **Kapla Meier** – then enter time period variable in **Time**, then Status of event occurrence in **Status & Define Event** such as 1 for event occurrence (death).
- Then enter factor variable in **Factor** to perform log rank test. At the top right of the interface opt for **Compare Factor -> Test Statistics -> Log rank** , finally click **Continue**.
- At the bottom select **Options** -> **Statistics** -> select **Survival table, Mean and Median Survival, Quartiles** . **Plots** -> **Survival.** At the end of the process click **ok.**

Means and Medians for Survival Time

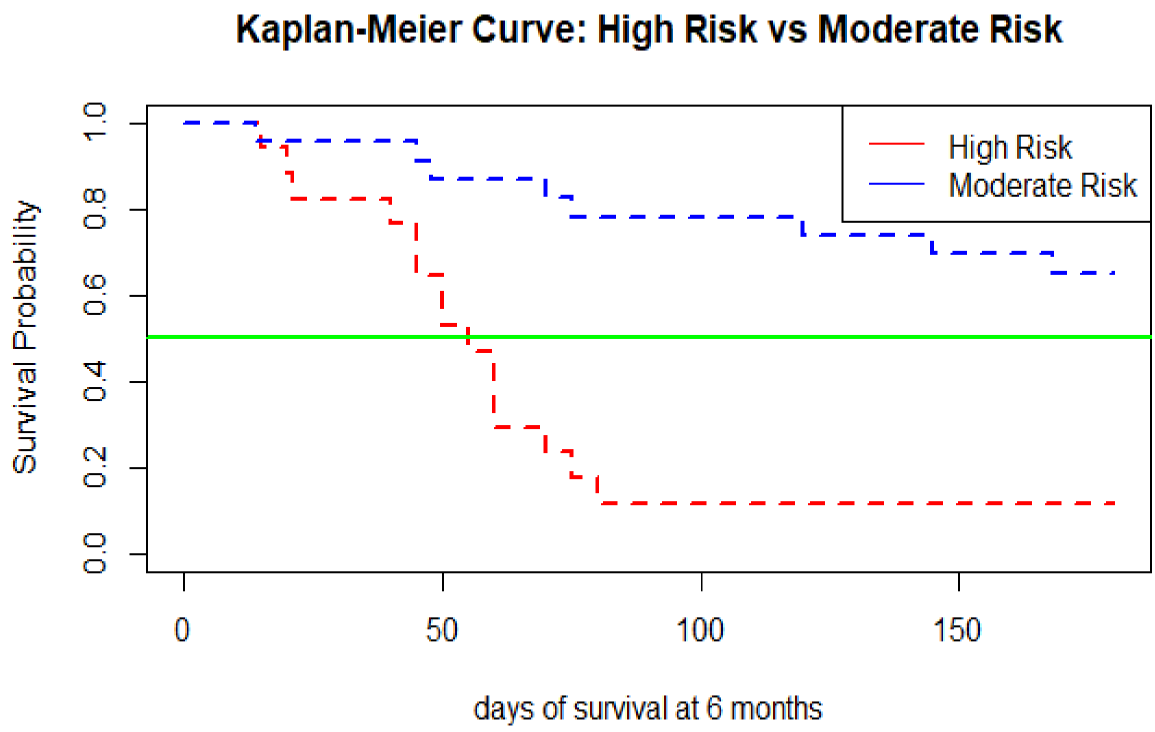
RISK_CATEGORY	Mean				Median			
	Estimate	Std. Error	95% Confidence Interval		Estimate	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound			Lower Bound	Upper Bound
Moderate Risk	147.174	11.213	125.196	169.152	.	.	.	.
High Risk	65.059	11.054	43.394	86.724	55.000	5.145	44.916	65.084
Overall	112.275	10.239	92.207	132.343	80.000	47.434	.000	172.971

a. Estimation is limited to the largest survival time if it is censored.

Overall Comparisons

	Chi-Square	df	Sig.
Log Rank (Mantel-Cox)	16.309	1	.000

Test of equality of survival distributions for the different levels of RISK_CATEGORY.

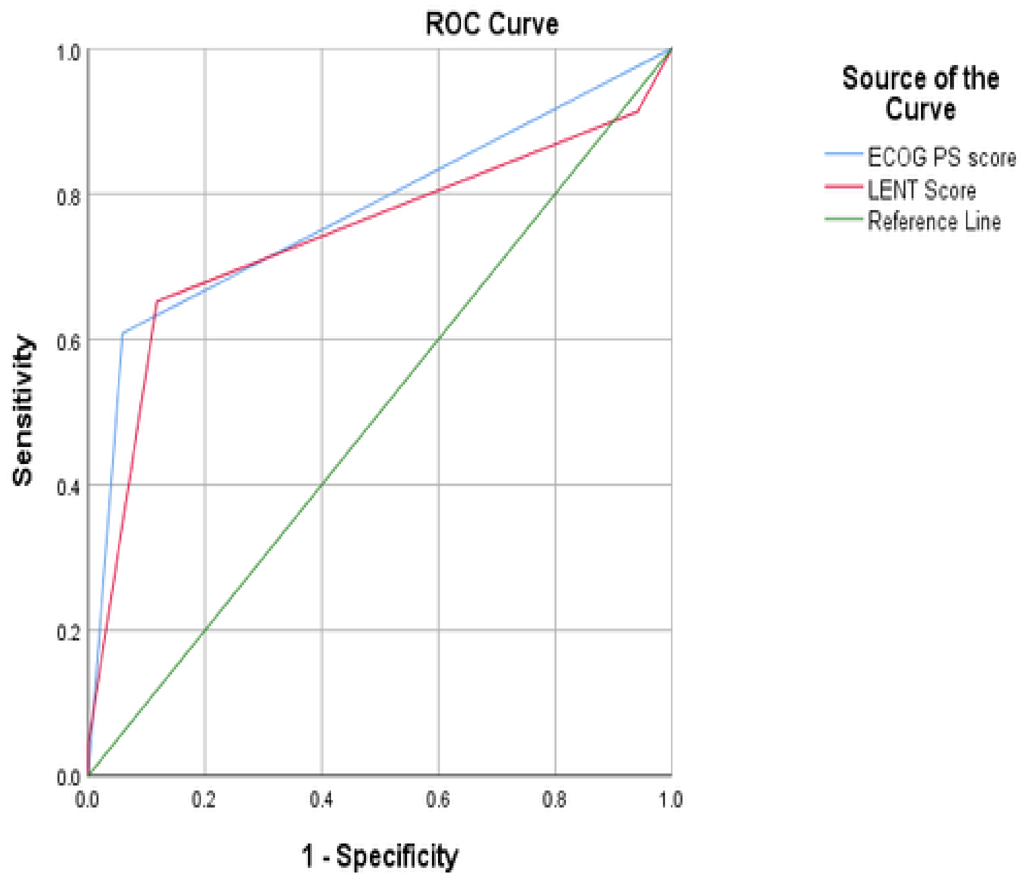


CONCLUSION :

- **From the plot we can conclude that the median survival time for high risk category patient as near to 55 days. Whereas it was observed that the average days of survival for the patients with moderate risk are 147.**
- **From the p value (< 0.05) of Log Rank test of equality of survival distributions for different risk category, by rejecting null hypothesis at 5% l.o.s we conclude that survival distribution are not same for both the risk categories.**

Steps to get Receiver Operator Curve in SPSS –

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **ROC Curve** – then enter **6 monts** in **State Variable**, & **ECOG PS score , LENT Score in Test Variable.**
- In **Display** section s**elect ROC Curve, with Diagonal reference line** and click **OK.**



Diagonal segments are produced by ties.

Conclusion:

- **ECOG PS score (AUC = 0.775): With this Area Under the Curve value, ECOG PS demonstrates good discriminatory ability. An AUC of 0.775 suggests that there's a 77.5% chance that the model will correctly distinguish a positive case from a negative one. This makes ECOG PS a fairly strong predictor in your context.**
- **LENT score (AUC = 0.742): While slightly lower, the LENT score still shows respectable predictive performance. An AUC of 0.742 means it's about 74.2% effective in classification, so it's useful, but not quite as strong as the ECOG PS score.**

Area Under the Curve					
Test Result Variable(s)	Area	Std. Error[a]	Asymptotic Sig.[b]	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
ECOG PS score	.775	.075	.003	.628	.922
LENT Score	.742	.081	.010	.583	.900
The test result variable(s): ECOG PS score, LENT Score has at least one tie between the positive actual state group and the negative actual state group. Statistics may be biased.					
a. Under the nonparametric assumption					
b. Null hypothesis: true area = 0.5					

Coordinates of the Curve			
Test Result Variable(s)	**Positive if Greater Than or Equal To[a]**	**Sensitivity**	**1 - Specificity**
ECOG PS score	.00	1.000	1.000
	1.50	.609	.059
	3.00	.000	.000
LENT Score	2.00	1.000	1.000
	3.50	.913	.941
	4.50	.652	.118
	5.50	.043	.000
	7.00	.000	.000
The test result variable(s): ECOG PS score, LENT Score has at least one tie between the positive actual state group and the negative actual state group.			

a. The smallest cutoff value is the minimum observed test value minus 1, and the largest cutoff value is the maximum observed test value plus 1. All the other cutoff values are the averages of two consecutive ordered observed test values.

Conclusion:

- **From the table Area Under The Curve we can observe that AUC for both ECOG PS score & LENT score is 0.775 & 0.742 respectively.**
- **P value for both the scores is less than 0.05 , this imply that we reject .**

H0: true AUC = 0.5 . and the model distinguish between events in a good manner.

Both scores are valuable, but ECOG PS seems to have a slight edge in performance for this specific task. If clinical decisions hinge on model accuracy, ECOG PS might be the preferable metric—though other factors like ease of use, data availability, and context-specific relevance also weigh in.

Dr. Athira Masterchart Data Analysis

DatasetLink –
https://docs.google.com/spreadsheets/d/1P30FXAoA5EnNisPMnWfk5GPEa_JyOT08/edit?usp=drive_link&ouid=102361808035996695223&rtpof=true&sd=true

CHI-square Test of Independence & Obtaining Column Percentage -

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data -> select **Count & Column Percent** from **Statistics** -> **Apply to Selection/ Apply to ALL** & click ok.
- P value obtained for all categorical study variables such as, Select **Analyse** -> **Descriptive Statistics** -> **Crosstabs** ->enter **HGS_GRADE** in **Columns(s)** and insert variabls [ASA,GENDER,T2DM,COPD,HTN,BLD_LOSS,LOS] in **Row(s)** -> at the right of the table select **Statistics** -> tick on **Chi-Square** & click OK.

Chisq2 test for association		HGS_GRADE				p value
		1		2		
		n	n %	n	n %	
ASA	1	5	8.20%	7	20.59%	
	2	32	52.46%	19	55.88%	
	3	23	37.70%	8	23.53%	
	4	1	1.64%	0	0.00%	0.203
GENDER	M	32	52.46%	16	47.06%	
	F	29	47.54%	18	52.94%	0.614
T2DM	0	23	37.70%	13	38.24%	
	1	38	62.30%	21	61.76%	0.959
COPD	0	41	67.21%	28	82.35%	
	1	20	32.79%	6	17.65%	0.113
HTN	0	10	16.39%	11	32.35%	
	1	51	83.61%	23	67.65%	0.072
BLD_LOSS	A	40	65.57%	25	73.53%	
	B	20	32.79%	9	26.47%	
	C	1	1.64%	0	0.00%	0.593
LOS	1	45	73.77%	29	85.29%	0.194

Conclusion :

- **From the above table it is found that, there is no statistically significant association exist between study variables and grouping vriable HGS_GRADE, as p value < 0.05.**

Comparing Means for Independent Groups -

- Open **SPSS** tab from desktop.
- Import **excel file**.
- Select **Analyze** -> **Tables** –> **Custom Tables** – then select grouping variable in **Columns** – then study variables in Rows.
- At the bottom left corner from **Define** opt for required **Summary Statistics** for quantitative data & **Categories and Totals** for categorical data & click ok.
- P value obtained for **BMI,AGE,CREAT,CRP,ALUBMIN** such as, Select **Analyse** -> **Compare Means** -> **Independent-Sample T Test** -> select **HGS_GRADE** as **Grouping Variable** Define Groups as per encoding done & Click ok.

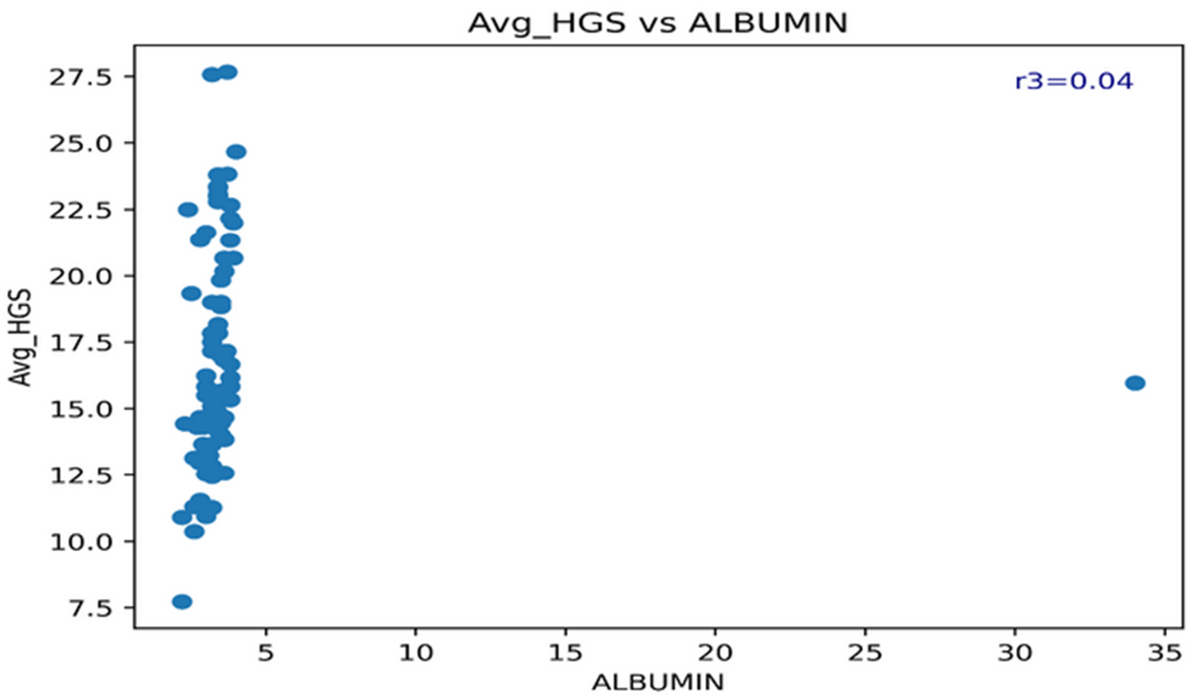
Descriptive Analysis /Independent t-test	**HGS_GRADE**				**t stat**	**P_value**
	1		**2**			
	Mean	**SD**	**Mean**	**SD**		
BMI	26.94	3.18	27.14	3.28	-0.292	0.77
AGE	73.51	7.27	71.74	4.52	1.287	0.2
CREAT	1.23	0.63	0.98	0.47	1.997	0.049
CRP	12.67	6.54	9.56	3.45	2.578	0.012
ALBUMIN	3.67	3.97	3.44	0.33	0.339	0.735

Conclusion :

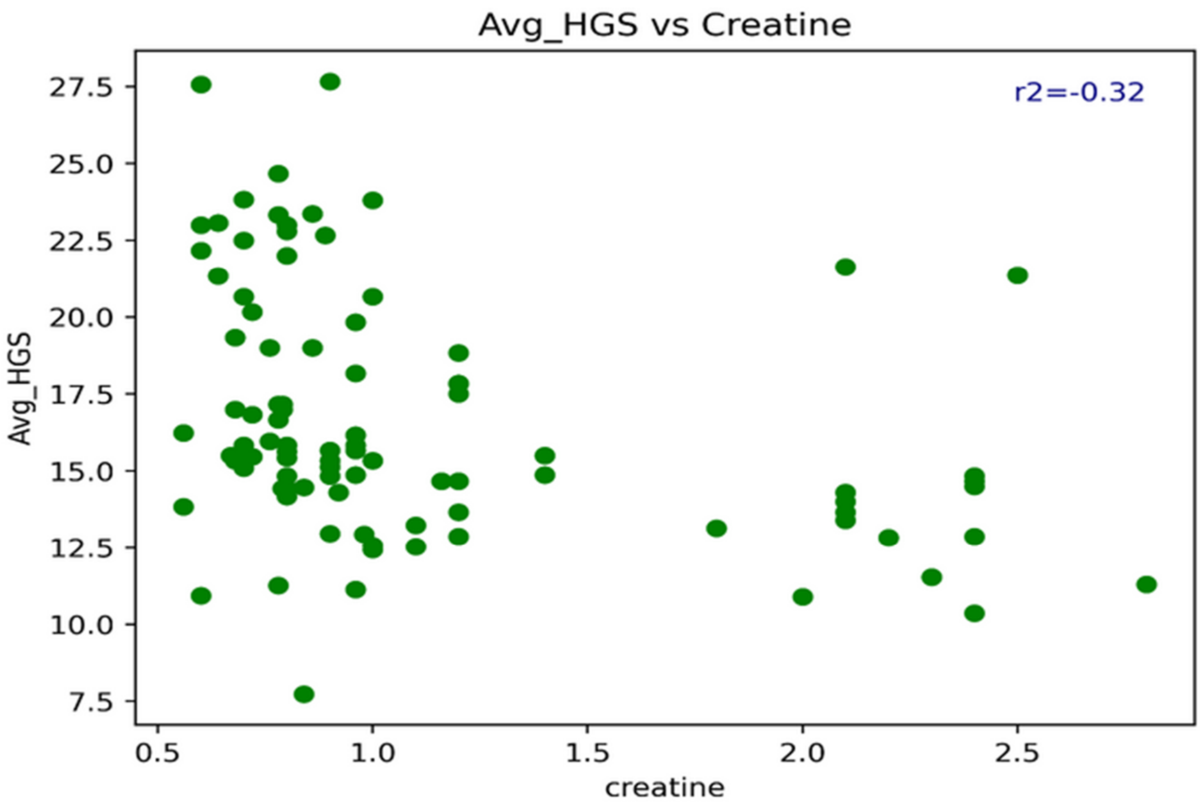
- **From the above table we can conclude that the average CREATININE and average CRP in both the groups of patients having HGS_GRADE 1 & 2 respectively is same at 5% l.O.S.**

Correlation Plots Obtained :

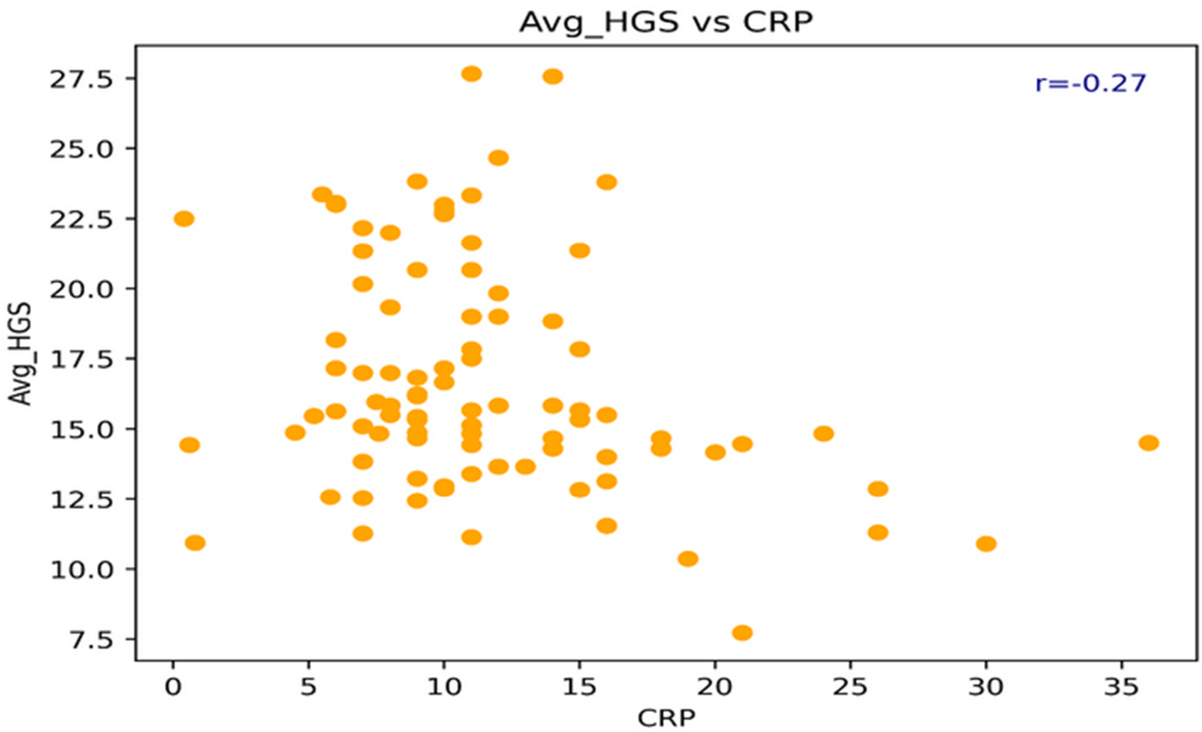
1.AVG_HGS vs ALBUMIN :



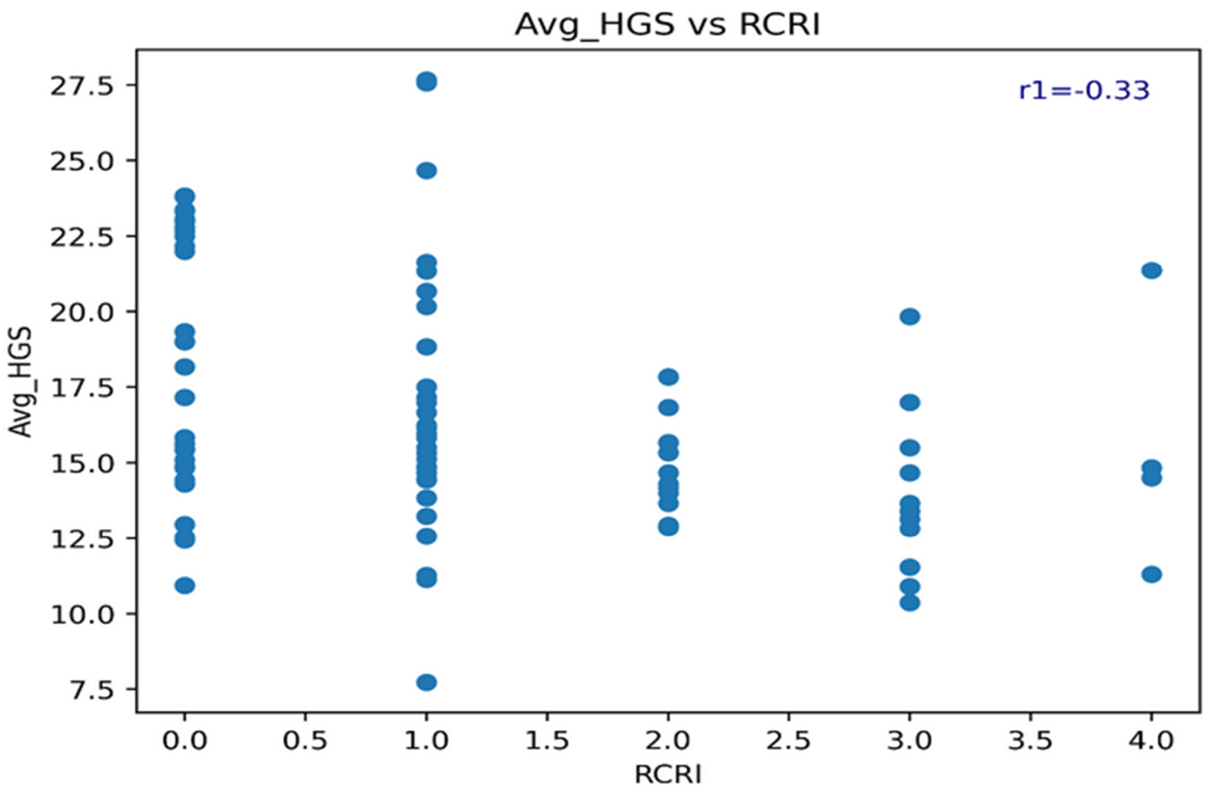
2.AVG_HGS vs CREATININE :



3.AVG_HGS vs CRP :



4.AVG_HGS vs RCRI :



From the above correlation plots it is observed that , there is no significant correlation exist between AVG_HGS with respect to ALBUMIN, CRETININE, CRP & RCRI. the strength of correlation is below moderate negative correlation.

12 SUMMARY

This report presents a comprehensive overview of the academic internship and training period undertaken at Bharti Hospital and Research Centre, with active involvement in the Central Research and Publication Unit (CRPU). The experience provided a valuable opportunity to explore the practical applications of biostatistics, medical research methods, data analysis tools, and modern machine learning techniques in the context of health sciences.

Throughout the internship, various fundamental and advanced concepts were studied, such as sampling techniques, study designs in medical research (including case-control, cohort, and RCTs), and statistical software applications like SPSS and SAS. The hands-on sessions significantly improved the ability to conduct data-driven investigations and interpret analytical results.

A highlight of the training was the dive into machine learning methods, including logistic regression, decision trees, random forests, and principal component analysis (PCA). These sessions not only enhanced technical skills but also emphasized the importance of feature selection and model evaluation in real-world medical datasets.

Special emphasis was placed on survival analysis and its importance in longitudinal medical studies. The internship also included guided assignments and group-based tasks that promoted collaborative learning and practical thinking.

In conclusion, this internship was a highly enriching academic journey that bridged theoretical knowledge with practical implementation. It not only enhanced statistical and analytical proficiency but also developed a strong foundation for conducting independent health research in the future.